

Exercise Set 10

MA2071

(1) Evaluate the surface integral $\int \int_S y \, dS$ where S is the surface $z = x + y^2$ above the rectangle $[0, 1] \times [0, 2]$.

(2) Evaluate the surface integral $\int \int_S x^2 y z \, dS$ where S is part of the plane $z = 1 + 2x + 3y$ that lies above the rectangle $[0, 3] \times [0, 2]$.

(3) Compute the flux of the vector field $F(x, y, z) = xy \mathbf{i} + yz \mathbf{j} + zx \mathbf{k}$ across S , where S is the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the square $[0, 1] \times [0, 1]$, and has upward orientation.

(4) Evaluate the line integral $\int_C F \, ds$ where $F(x, y, z) = (x + y^2) \mathbf{i} + (y + z^2) \mathbf{j} + (z + x^2) \mathbf{k}$ and C is the triangle with vertices $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.

(5) Evaluate $\int \int_S \text{curl } F \cdot N \, dS$ where $F(x, y, z) = yz \mathbf{i} + xz \mathbf{j} + xy \mathbf{k}$ and S is the part of the paraboloid $z = 9 - x^2 - y^2$ that lies above the plane $z = 5$, oriented upwards.

(6) Calculate the flux of the vector field $F(x, y, z) = 3xy^2 \mathbf{i} + xe^z \mathbf{j} + z^3 \mathbf{k}$ across the surface S , where S is the surface of the solid bounded by the cylinder $y^2 + z^2 = 1$ and the planes $x = -1$ and $x = 2$.

(7) Calculate the flux of the vector field $F(x, y, z) = (\cos z + xy^2) \mathbf{i} + xe^{-z} \mathbf{j} + (\sin y + x^2 z) \mathbf{k}$ across the surface S , where S is the surface of the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.